

Understanding Analysis, 2nd Edition

Chapter 5 - Section 5.2 - Derivatives and the Intermediate Value Property

1. Supply proofs for part (i) and (ii) of Theorem 5.2.4.

Solution: Part (i): $(f + g)'(c) = f'(c) + g'(c)$

Proof:

$$\begin{aligned}(f + g)'(c) &= \lim_{x \rightarrow c} \frac{f(x) + g(x) - f(c) - g(c)}{x - c} = \lim_{x \rightarrow c} \left(\frac{f(x) - f(c)}{x - c} + \frac{g(x) - g(c)}{x - c} \right) \\ &= f'(c) + g'(c)\end{aligned}$$

where the last step is justified because we know both limits exist, so we can apply the Algebraic Limit Theorem (Theorem 4.2.4).

Part (ii): $(kf)'(c) = kf'(c)$ for all $k \in \mathbf{R}$.

Proof:

$$(kf)'(c) = \lim_{x \rightarrow c} k \left(\frac{f(x) - f(c)}{x - c} \right) = k \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = kf'(c)$$

2. Exactly one of the following requests is impossible. Decide which it is, and provide examples for the other three. In each case, assume the functions are defined on all of $\mathbf{R}$.

- (a) Functions f and g not differentiable at zero but where fg is differentiable at zero.
- (b) A function f not differentiable at zero and a function g differentiable at zero where fg is differentiable at zero.
- (c) A function f not differentiable at zero and a function g differentiable at zero where $f + g$ is differentiable at zero.
- (d) A function f differentiable at zero but not differentiable at any other point.

Solution:

- (a) $f(x) = 0$ for $x \geq 0$ and $f(x) = 1$ otherwise. $g(x) = 1$ for $x \geq 0$ and $g(x) = 0$ otherwise. Both functions are discontinuous at zero hence not differentiable there, but fg is the zero function.

- (b) $f(x) = 0$ for $x \geq 0$ and $f(x) = 1$ otherwise. $g(x) = 0$ everywhere.
- (c) Impossible. If $f + g$ were differentiable at zero, then $(f + g) - g = f$ must be differentiable at zero as well by Theorem 5.2.4.
- (d) $f(x) = x^2$ for $x \in \mathbf{Q}$ and $f(x) = 0$ otherwise.

It's differentiable at zero because for $c = 0$, $\frac{f(x)-f(c)}{x-c} = x$ and $\lim_{x \rightarrow 0} x = 0$.

It's not differentiable at $c \neq 0$. Consider a sequence $(x_n) \rightarrow c$. We can split it into two subsequences $(a_n) \in \mathbf{Q}$ and $(b_n) \notin \mathbf{Q}$. We have $(a_n) \rightarrow c^2$ and $(b_n) \rightarrow 0$. Since $c^2 \neq 0$ for $c \neq 0$, the two subsequences converge to different limits, hence (x_n) doesn't converge.

3. (a) Use Definition 5.2.1 to produce the proper formula for the derivation of $f(x) = 1/x$.
- (b) Combine the result in part (a) with the Chain Rule to supply a proof for part (iv) of Theorem 5.2.4.
- (c) Supply a direct proof of Theorem 5.2.4 (iv) by algebraically manipulating the difference quotient for (f/g) in a style similar to the proof of Theorem 5.2.4 (iii).

Solution:

(a)

$$\frac{f(x) - f(c)}{x - c} = \frac{\frac{1}{x} - \frac{1}{c}}{x - c} = \frac{-(x - c)}{xc} \cdot \frac{1}{x - c} = -\frac{1}{xc}$$

So we have $f'(c) = \lim_{x \rightarrow c} (-\frac{1}{xc}) = -\frac{1}{c^2}$ for $c \neq 0$.

(b) We first derive a formula for the derivative of $h(x) = 1/g(x)$.

$$\frac{h(x) - h(c)}{x - c} = \frac{-(g(x) - g(c))}{g(x)g(c)} \cdot \frac{1}{x - c} = -\frac{1}{g(x)g(c)} \cdot \frac{g(x) - g(c)}{x - c}$$

Taking the limit to c and noting that the second term of the product is the definition of $g'(c)$

$$h'(c) = \lim_{x \rightarrow c} -\frac{1}{g(x)g(c)} \cdot \frac{g(x) - g(c)}{x - c} = -\frac{g'(c)}{g(c)^2}$$

Now apply the product rule to $(fh)'(c)$ to get the formula for $(f/g)'(c)$.

— Simpler solution by chatgpt —

Derive the formula of $h'(x)$ by defining $h(x) = 1/x$ then apply the chain rule to $h(g(x))$.

(c)

$$\begin{aligned}\frac{\frac{f}{g}(x) - \frac{f}{g}(c)}{x - c} &= \frac{\frac{f}{g}(x) - \frac{f(x)}{g(c)} + \frac{f(x)}{g(c)} - \frac{f}{g}(c)}{x - c} \\&= \frac{f(x)}{x - c} \left(\frac{g(c) - g(x)}{g(x)g(c)} \right) + \frac{1}{g(c)(x - c)} (f(x) - f(c)) \\&= -\frac{f(x)}{g(x)g(c)} \left(\frac{g(x) - g(c)}{x - c} \right) + \frac{1}{g(c)} \left(\frac{f(x) - f(c)}{x - c} \right)\end{aligned}$$

Take the limit to c to get

$$-\frac{f(c)g'(c)}{g(c)^2} + \frac{f'(c)}{g(c)} = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$$

4. Follow these steps to provide a slightly modified proof of the Chain Rule.

- (a) Show that a function $h : A \rightarrow \mathbf{R}$ is differentiable at $a \in A$ if and only if there exists a function $l : A \rightarrow \mathbf{R}$ which is continuous at a and satisfies

$$h(x) - h(a) = l(x)(x - a) \quad \text{for all } x \in A.$$

- (b) Use this criterion for differentiability (in both directions) to prove Theorem 5.2.5.

Solution:

- (a) First we prove that if such a function l exists then h is differentiable at a . We have

$$h'(a) = \lim_{x \rightarrow a} \frac{h(x) - h(a)}{x - a} = \lim_{x \rightarrow a} l(x) = l(a)$$

Next, we prove that if h is differentiable at a then such a function l exists. Define

$$l(x) = \begin{cases} \frac{h(x) - h(a)}{x - a} & x \neq a \\ h'(a) & (x = a) \end{cases}$$

l is continuous at a since $\lim_{x \rightarrow a} l(x) = \lim_{x \rightarrow a} \frac{h(x) - h(a)}{x - a} = h'(a) = l(a)$.

$h(x) - h(a) = l(x)(x - a)$ for all $x \in A$ by definition.

- (b) TODO.

5. Let $l(x) = \begin{cases} x^a & x > 0 \\ 0 & x \leq 0 \end{cases}$.

- (a) For which values of a is f continuous at zero?
- (b) For which values of a is f differentiable at zero? In this case, is the derivative function continuous?
- (c) For which values of a is f twice-differentiable?

Solution:

(a) $a > 0$. The limit at 0 from the left is 0. If $a < 0$ then the right limit diverges. If $a = 0$ then the right limit is 1. But if $a > 0$ then the right limit is 0.

(b) From the direction of positive x , we have $\lim_{x \rightarrow 0^+} \frac{x^a - 0^a}{x - 0} = \lim_{x \rightarrow 0^+} x^{a-1} = 0^{a-1}$. This is only defined for $a > 1$, in which case it's 0 and agrees with the left limit, so the derivative is defined.

The derivative is continuous, but seems to require us to use the fact that $l'(x) = ax^{a-1}$ for *non-integer* a , which we have not proved yet.

(c) The next right-side derivative at zero is $\lim_{x \rightarrow 0^+} x^{a-2} = 0^{a-2}$, so we need $a > 2$.

6. Let g be defined on an interval A and let $c \in A$.

- (a) Explain why $g'(c)$ in Definition 5.2.1 could have been given by

$$g'(c) = \lim_{h \rightarrow 0} \frac{g(c+h) - g(c)}{h}$$

- (b) Assume A is open. If g is differentiable at $c \in A$, show

$$g'(c) = \lim_{h \rightarrow 0} \frac{g(c+h) - g(c-h)}{2h}$$

Solution:

- (a) Let $h = x - c$. Then we have

$$g'(c) = \lim_{(x-c) \rightarrow 0} \frac{g(x) - g(c)}{x - c}$$

And use the fact that $\lim_{(x-c) \rightarrow 0} = \lim_{x \rightarrow c}$.

(b)

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{g(c+h) - g(c-h)}{2h} &= \lim_{h \rightarrow 0} \frac{g(c+h) - g(c) + g(c) - g(c-h)}{2h} \\ &= \frac{1}{2} \lim_{h \rightarrow 0} \left(\frac{g(c+h) - g(c)}{h} + \frac{g(c) - g(c-h)}{h} \right) \\ &= \frac{1}{2} (g'(c) + g'(c)) = g'(c)\end{aligned}$$

We used the fact that $g'(c) = \lim_{h \rightarrow 0} \frac{g(c) - g(c-h)}{h}$, which we'll now prove. Let $h = c - x$. Then we have

$$\lim_{(c-x) \rightarrow 0} \frac{g(c) - g(x)}{c - x} = \lim_{(c-x) \rightarrow 0} \frac{g(x) - g(c)}{x - c}$$

And use the fact that $\lim_{(c-x) \rightarrow 0} = \lim_{(x-c) \rightarrow 0} = \lim_{x \rightarrow c}$ (this is easy to prove with the usual $\epsilon - \delta$).

One more thing to note is why the problem assumes A is open. This is so that both $g(c-h)$ and $g(c+h)$ are defined.

— Slight improvement by chatgpt —

To prove $g'(c) = \lim_{h \rightarrow 0} \frac{g(c) - g(c-h)}{h}$, set $k = -h$. Then as $h \rightarrow 0$, also $k \rightarrow 0$, and

$$\frac{g(c) - g(c-h)}{h} = \frac{g(c+k) - g(c)}{k}$$

Then reuse part (a).

7. Let

$$g_a(x) = \begin{cases} x^a \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Find a particular (potentially noninteger) value for a so that

- (a) g_a is differentiable on $\mathbf{R}$ but such that g'_a is unbounded on $[0, 1]$.
- (b) g_a is differentiable on $\mathbf{R}$ with g'_a continuous but not differentiable at zero.
- (c) g_a is differentiable on $\mathbf{R}$ and g'_a is differentiable on $\mathbf{R}$, but such that g''_a is not continuous at zero.

Solution: We're going to use the fact that the derivative of x^a is ax^{a-1} for $a \in \mathbf{R}$, even though the book has only proved it for $a \in \mathbf{N}$.

(a)

$$g'_a(x) = \begin{cases} ax^{a-1} \sin(1/x) - x^{a-2} \cos(1/x) & \text{if } x \neq 0 \\ \lim_{x \rightarrow 0} x^{a-1} \sin(1/x) & \text{if } x = 0 \end{cases}$$

For $a < 1$, $g'(0)$ is not defined because the x^{a-1} makes it unbounded. For $a = 1$, $g'(0) = \lim_{x \rightarrow 0} \sin(1/x)$, which is also not defined (Example 4.2.6). For $a \geq 2$, $g'_a(x)$ is bounded in $[0, 1]$.

The answer is then $1 < a < 2$, for which $g'_a(x)$ is defined everywhere and, by the $x^{a-2} \cos(1/x)$ term is unbounded as $x \rightarrow 0$. But we need to be a little careful in choosing a to avoid even roots like $\sqrt{-1}$ which is not defined in $\mathbf{R}$. We can choose $a = \frac{5}{3}$, which gives

$$g'_a(x) = \begin{cases} \frac{5}{3}x^{2/3} \sin(1/x) - x^{-\frac{1}{3}} \cos(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

(b) Let $a = 3$. Then

$$g'_a(x) = \begin{cases} 3x^2 \sin(1/x) - x \cos(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Which is continuous at zero. But the second derivative at zero

$$g''_a(0) = \lim_{x \rightarrow 0} \frac{g'(x) - g'(0)}{x} = \lim_{x \rightarrow 0} 3x \sin(1/x) - \cos(1/x)$$

is undefined, since $3x \sin(1/x)$ tends to zero but $\cos(1/x)$ oscillates between -1 and 1.

(c) Let $a = 4$. Then

$$g'_a(x) = \begin{cases} 4x^3 \sin(1/x) - x^2 \cos(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

And the second derivative

$$g''_a(x) = \begin{cases} 12x^2 \sin(1/x) - 6x \cos(1/x) - \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Which is discontinuous at zero because in the equation for $x \neq 0$, the first two terms tend to zero while $\sin(1/x)$ oscillates between -1 and 1.

8. TODO

9. True / false with proof / counterexample.

- (a) If f' exists on an interval and is not constant, then f' must take on some irrational values.
- (b) If f' exists on an open interval and there is some point c where $f'(c) > 0$, then there exists a δ -neighborhood $V_\delta(c)$ around c in which $f'(x) > 0$ for all $x \in V_\delta(c)$.
- (c) If f is differentiable on an interval containing zero and if $\lim_{x \rightarrow 0} f'(x) = L$, then it must be that $L = f'(0)$.

Solution:

(a) True, by Darboux Theorem and the fact that the irrationals are dense in $\mathbf{R}$.

(b) — **Incorrect** —

True. Otherwise $f'(c)$ is an isolated point and that violates Darboux Theorem.

— **Corrected** —

The above is wrong because Darboux Theorem only guarantees Intermediate Value Property and not continuity (see Section 4.5).

I would not have been able to come up with the counterexample myself. It's

$$f(x) = \begin{cases} x/2 + x^2 \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

The $x/2$ term makes $f'(0) = \frac{1}{2}$ (compare this with $f(x) = x^2 + \sin(1/x)$ where $f'(0)$ would be 0).

For $x \neq 0$, $f'(x) = \frac{1}{2} - \cos(1/x) + 2x \sin(1/x)$. For any $V_\delta(0)$ there's always an x such that $\cos(1/x) = 1$, which would make $f'(x)$ negative.

(c) — **Incorrect** —

We need the following lemma: if $\lim_{a \rightarrow b} f(a)$ is defined for some b near c , and $\lim_{a \rightarrow c} f(a)$ is also defined, then $\lim_{b \rightarrow c} (\lim_{a \rightarrow b} f(a)) = \lim_{a \rightarrow c} f(a)$.

The proof is straightforward with $\epsilon - \delta$. There's a δ such that $|a - c| < \delta \implies |f(a) - L| < \epsilon$. Make $|b - c| < \delta/2$ and $|a - b| < \delta/2$.

It's very important to note that $\lim_{a \rightarrow b} f(a)$ *must* be defined for b near c for this lemma to work. Otherwise, one counterexample is the Dirichlet function.

$\lim_{a \rightarrow 0} f(a) = 0$. But the limit is undefined for all $b \neq 0$ so the expression $\lim_{b \rightarrow 0}(\lim_{a \rightarrow b} f(a))$ is undefined.

Now back to the original problem. f is differentiable on an interval containing zero, so $f'(0)$ exists and also

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists for c near 0.

We also have

$$\lim_{c \rightarrow 0} \left(\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right) = L$$

Now apply the lemma to conclude that

$$\lim_{c \rightarrow 0} \left(\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = f'(0) = L$$

— **Corrected** —

The nested limit lemma above only applies to functions with a single parameter. For a counterexample, consider $\lim_{c \rightarrow 0}(\lim_{x \rightarrow c} \frac{x}{c}) = \lim_{c \rightarrow 0} 1 = 1$. But $\lim_{x \rightarrow 0} \frac{x}{c}$ is undefined.

Instead we can use Darboux Theorem. Suppose for contradiction that $f'(0) \neq L$ and let $d = |f'(0) - L|$. By convergence to L , there exists δ such that $x \in V_\delta(0) \implies |f'(x) - L| < d/2$.

But by Darboux, for such $x \in V_\delta(0)$ then the derivative must take every value between $f'(x)$ and $f'(0)$. In particular, there exists $y \in V_\delta(0)$ such that $|f'(y) - f'(0)| < d/2$, contradicting convergence to L ($f'(y)$ cannot be within $d/2$ of both L and $f'(0)$).

10. Recall that a function $f : (a, b) \rightarrow \mathbf{R}$ is increasing on (a, b) if $f(x) \leq f(y)$ whenever $x < y$ in (a, b) . A familiar mantra from calculus is that a differentiable function is increasing if its derivative is positive, but this statement requires some sharpening in order to be completely accurate. Show that the function

$$g(x) = \begin{cases} x/2 + x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is differentiable on $\mathbf{R}$ and satisfies $g'(0) > 0$. Now, prove that g is *not* increasing over any open interval containing 0.

In the next section we will see that f is indeed increasing on (a, b) if and only if $f'(x) \geq 0$ for all $x \in (a, b)$.

Solution: Heh, turns out this is the solution to the previous problem... Anyway, the derivative is

$$g'(x) = \begin{cases} \frac{1}{2} + 2x \sin(1/x) - \cos(1/x) & \text{if } x \neq 0 \\ \frac{1}{2} & \text{if } x = 0 \end{cases}$$

So g is differentiable on $\mathbf{R}$ and $g'(0) > 0$ as required.

To prove that g is not increasing over any open interval containing 0, choose an x such that $\cos(1/x) = 1$, hence $\sin(1/x) = 0$ and $g'(x) = -\frac{1}{2}$. Since the derivative is negative, g cannot be increasing.

11. Assume that g is differentiable on $[a, b]$ and satisfies $g'(a) < 0 < g'(b)$.
- (a) Show that there exists a point $x \in (a, b)$ where $g(a) > g(x)$ and a point $y \in (a, b)$ where $g(y) < g(b)$.
- (b) Now complete the proof of Darboux Theorem started earlier.

Solution:

- (a) The derivative at a is

$$\lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h} < 0$$

The denominator is positive. For the result to be negative, it must be that $g(a+h) < g(a)$ for small enough h .

Similarly, the derivative at b is

$$\lim_{h \rightarrow 0} \frac{g(b) - g(b-h)}{h} > 0$$

The denominator is positive. For the result to be positive, it must be that $g(b) > g(b-h)$ for small enough h .

- (b) To recap the book's partial proof of the theorem: let $g(x) = f(x) - \alpha x$ on $[a, b]$. Notice that g is differentiable on $[a, b]$ with $g'(x) = f'(x) - \alpha$. In terms of g , the theorem's hypothesis states that $g'(a) < 0 < g'(b)$ and we need to show that $g'(c) = 0$ for some $c \in [a, b]$.

Now to complete the proof. g differentiable on $[a, b]$ implies it's continuous on $[a, b]$. By the Extreme Value Theorem, g attains a minimum value at a point c . By part(a), $c \neq a$ and $c \neq b$. By the Interior Extremum Theorem, $g'(c) = 0$.

12. **(Inverse functions).** If $f : [a, b] \rightarrow \mathbf{R}$ is one-to-one, then there exists an inverse function f^{-1} defined on the range of f given by $f^{-1}(y) = x$ where $y = f(x)$. In Exercise 4.5.8 we saw that if f is continuous on $[a, b]$, then f^{-1} is continuous on its domain. Let's add the assumption that f is differentiable on $[a, b]$ with $f'(x) \neq 0$ for all $x \in [a, b]$. Show f^{-1} is differentiable with

$$(f^{-1})'(y) = \frac{1}{f'(x)}$$

Solution: Let $c \in [a, b]$. Then the derivative of f^{-1} at $f(c)$ is

$$\begin{aligned} (f^{-1})'(f(c)) &= \lim_{y \rightarrow f(c)} \frac{f^{-1}(y) - f^{-1}(f(c))}{y - f(c)} \\ &= \lim_{y \rightarrow f(c)} \frac{x - c}{f(x) - f(c)} \end{aligned}$$

Since f^{-1} is continuous on its domain, we can conclude that $y \rightarrow f(c) \implies f^{-1}(y) \rightarrow f^{-1}(f(c))$ or equivalently, $y \rightarrow f(c) \implies x \rightarrow c$. Applying this to the limit above

$$\begin{aligned} (f^{-1})'(f(c)) &= \lim_{y \rightarrow f(c)} \frac{x - c}{f(x) - f(c)} \\ &= \lim_{x \rightarrow c} \frac{x - c}{f(x) - f(c)} \\ &= \frac{1}{f'(c)} \end{aligned}$$

Note that in the second equality, $f(x) - f(c) \neq 0$ in the denominator because f is one-to-one, hence $x \neq c \implies f(x) \neq f(c)$. In the last equality, $f'(c) \neq 0$ by the problem's assumption.